

Lab 1

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Part 1: Capturing HTTP Traffic

Task 2 – HTTP request/response captured for neverssl.com

		Info	length	Protocol	Destination	Source	Time	No.
0000	bc 3d 85 d4 78 7	GET /online/ HTTP/1.1	567	HTTP	34.223.124.45	192.168.8.117	44.663764900	1989
0010	02 29 95 81 40 0	HTTP/1.1 200 OK (text/html)	300	HTTP	192.168.8.117	34.223.124.45	45.112375500	14043
0020	7c 2d e1 5c 00 5	GET /favicon.ico HTTP/1.1	476	HTTP	34.223.124.45	192.168.8.117	45.227211100	14079
0030	ff ff 6a 45 00 0	HTTP/1.1 200 OK (PNG)	470	HTTP	192.168.8.117	34.223.124.45	45.473649200	14137

0000	bc 3d 85 d4 78 7	13989: Packet, 567 bytes on wire (4536 bits), 567 bytes captured (4536 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD0D5105F}, id 0 <
0010	02 29 95 81 40 0	Ethernet II, Src: Intel_b6:5f:03 (a0:29:42:b6:5f:03), Dst: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74) <
0020	7c 2d e1 5c 00 5	Internet Protocol Version 4, Src: 192.168.8.117, Dst: 34.223.124.45 <
0030	ff ff 6a 45 00 0	Transmission Control Protocol, Src Port: 80, Seq: 1, Ack: 1, Len: 513 <
0040	65 2f 20 48 54 5	Hypertext Transfer Protocol >
0050	74 3a 20 6e 65 7	GET /online/ HTTP/1.1\r\n
0060	0a 43 6f 6e 66 6	Host: neverssl.com\r\n
0070	70 2d 61 6c 69 7	Connection: keep-alive\r\n
0080	2d 49 6e 73 65 6	Upgrade-Insecure-Requests: 1\r\n
0090	74 73 3a 20 31 0	User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36 Edg/153.0.0.0\r\n
00a0	74 3a 20 4d 6f 7	Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
00b0	57 69 6e 64 6f 7	Referer: https://www.bing.com/\r\n
00c0	20 57 69 6e 36 3	Accept-Encoding: gzip, deflate\r\n
00d0	6c 65 57 65 62 4	Accept-Language: ar,en;q=0.9,en-GB;q=0.8,en-US;q=0.7,ar-SA;q=0.6\r\n
00e0	28 40 48 54 4d 4	\r\n
00f0	6b 6f 29 20 43 6	[Response in frame: 14043]
0100	2e 30 2e 30 20 5	
0110	33 36 20 45 64 6	
0120	0d 0a 41 63 63 6	

Part 2: Analyzing TCP/IP Traffic

Task 1 – Follow TCP Stream

Wireshark · Follow TCP Stream (tcp.stream eq 88) · شبكة Wi-Fi	
GET /online/ HTTP/1.1 Host: neverssl.com Connection: keep-alive Upgrade-Insecure-Requests: 1 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36 Edg/153.0.0.0 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 Referer: https://www.bing.com/ Accept-Encoding: gzip, deflate Accept-Language: ar,en;q=0.9,en-GB;q=0.8,en-US;q=0.7,ar-SA;q=0.6	
HTTP/1.1 200 OK Date: Tue, 22 Sep 2026 02:55:14 GMT Server: Apache/2.4.68 () Upgrade: h2,h2c Connection: Upgrade, Keep-Alive Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT ETag: "8be-5e28b29291e10-gzip" Accept-Ranges: bytes Vary: Accept-Encoding Content-Encoding: gzip Content-Length: 1173 Keep-Alive: timeout=5, max=100 Content-Type: text/html; charset=UTF-8	
.....uVQo.6~..{..X..,i.....Y.....(.T.\$J.L..IEU...Q.....@.y.....N...Q.8....zF;/....A...=\$P..J].`.:..... b.. c.....(.L1...J.)Rb#.....q h%.%8...cY...U'..-~.y.[.Q.n..../15h+.....\x...j....p...3....D.M ...u.eFjG N3.'..-6Hs.C:..e._.^.....t.pC...7..p.bQ....%.L.m;...+.hv.a~...t.....]......%D.b==.K.&.<....I.. ...5	

Task 2 (Step 1–2) – Three-Way Handshake: SYN

tcp.port==57692

Info	enpth	Protocol	Destination	Source	Time	No
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 80 → 57692 66		TCP	34.223.124.45	192.168.8.117	44.383888700	13945
Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 [SYN, ACK] 57692 → 80 58		TCP	192.168.8.117	34.223.124.45	44.663418300	13987
Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	44.663514300	13988
GET /online/ HTTP/1.1 567		HTTP	34.223.124.45	192.168.8.117	44.663764900	13989
Seq=1 Ack=514 Win=26800 Len=0 [ACK] 57692 → 80 54		TCP	192.168.8.117	34.223.124.45	44.934746300	14017
Seq=1 Ack=514 Win=26800 Len=1300 [TCP PDU reassembled in 14043] [ACK] 57692 → 80 1354		TCP	192.168.8.117	34.223.124.45	45.111933000	14042
HTTP/1.1 200 OK (text/html) 300		HTTP	192.168.8.117	34.223.124.45	45.112375500	14043
Seq=514 Ack=1547 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	45.112430800	14044

rame 13945: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD005105F}, id 0 <
Ethernet II, Src: Intel_b6:5f:03 (a0:29:42:b6:5f:03), Dst: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74) <
Internet Protocol Version 4, Src: 192.168.8.117, Dst: 34.223.124.45 <
Transmission Control Protocol, Src Port: 57692, Dst Port: 80, Seq: 0, Len: 0 <
Source Port: 57692
Destination Port: 80
[Stream index: 88]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)] <
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3766063609
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
Header Length: 32 bytes (8) = 1000
Flags: 0x002 (SYN) <
Window: 65535
[Calculated window size: 65535]

Task 2 (Step 1–2) – Three-Way Handshake: SYN, ACK

tcp.port==57692

Info	enpth	Protocol	Destination	Source	Time	No
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 80 → 57692 66		TCP	34.223.124.45	192.168.8.117	44.383888700	13945
Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 [SYN, ACK] 57692 → 80 58		TCP	192.168.8.117	34.223.124.45	44.663418300	13987
Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	44.663514300	13988
GET /online/ HTTP/1.1 567		HTTP	34.223.124.45	192.168.8.117	44.663764900	13989
Seq=1 Ack=514 Win=26800 Len=0 [ACK] 57692 → 80 54		TCP	192.168.8.117	34.223.124.45	44.934746300	14017
Seq=1 Ack=514 Win=26800 Len=1300 [TCP PDU reassembled in 14043] [ACK] 57692 → 80 1354		TCP	192.168.8.117	34.223.124.45	45.111933000	14042
HTTP/1.1 200 OK (text/html) 300		HTTP	192.168.8.117	34.223.124.45	45.112375500	14043
Seq=514 Ack=1547 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	45.112430800	14044

rame 13987: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD005105F}, id 0 <
Ethernet II, Src: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74), Dst: Intel_b6:5f:03 (a0:29:42:b6:5f:03) <
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.8.117 <
Transmission Control Protocol, Src Port: 80, Dst Port: 57692, Seq: 0, Ack: 1, Len: 0 <
Source Port: 80
Destination Port: 57692
[Stream index: 88]
[Stream Packet Number: 2]
[Conversation completeness: Complete, WITH_DATA (31)] <
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 823472170
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3766063610
Header Length: 24 bytes (6) = 0110
Flags: 0x012 (SYN, ACK) <
Window: 26883
[Calculated window size: 26883]

Task 2 (Step 1–2) – Three-Way Handshake: ACK

tcp.port==57692

Info	enpth	Protocol	Destination	Source	Time	No
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 80 → 57692 66		TCP	34.223.124.45	192.168.8.117	44.383888700	13945
Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 [SYN, ACK] 57692 → 80 58		TCP	192.168.8.117	34.223.124.45	44.663418300	13987
Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	44.663514300	13988
GET /online/ HTTP/1.1 567		HTTP	34.223.124.45	192.168.8.117	44.663764900	13989
Seq=1 Ack=514 Win=26800 Len=0 [ACK] 57692 → 80 54		TCP	192.168.8.117	34.223.124.45	44.934746300	14017
Seq=1 Ack=514 Win=26800 Len=1300 [TCP PDU reassembled in 14043] [ACK] 57692 → 80 1354		TCP	192.168.8.117	34.223.124.45	45.111933000	14042
HTTP/1.1 200 OK (text/html) 300		HTTP	192.168.8.117	34.223.124.45	45.112375500	14043
Seq=514 Ack=1547 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	45.112430800	14044

rame 13988: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD005105F}, id 0 <
Ethernet II, Src: Intel_b6:5f:03 (a0:29:42:b6:5f:03), Dst: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74) <
Internet Protocol Version 4, Src: 192.168.8.117, Dst: 34.223.124.45 <
Transmission Control Protocol, Src Port: 57692, Dst Port: 80, Seq: 1, Ack: 1, Len: 0 <
Source Port: 57692
Destination Port: 80
[Stream index: 88]
[Stream Packet Number: 3]
[Conversation completeness: Complete, WITH_DATA (31)] <
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3766063610
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 823472171
Header Length: 20 bytes (5) = 0101
Flags: 0x010 (ACK) <
Window: 65535
[Calculated window size: 65535]

Task 2 (Step 3) – Data Transfer

		Info	length	Protocol	Destination	Source	Time	No.
		Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 80 → 57692 66		TCP	34.223.124.45	192.168.8.117	44.38388700	13945
		Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 [SYN, ACK] 57692 → 80 58		TCP	192.168.8.117	34.223.124.45	44.663418300	13987
		Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	44.663514300	13988
		GET /online/ HTTP/1.1 567		HTTP	34.223.124.45	192.168.8.117	44.663764900	13989
		Seq=1 Ack=514 Win=26800 Len=0 [ACK] 57692 → 80 54		TCP	192.168.8.117	34.223.124.45	44.934746300	14017
		Seq=1 Ack=514 Win=26800 Len=1300 [TCP PDU reassembled in 14043] [ACK] 57692 → 80 1354		TCP	192.168.8.117	34.223.124.45	45.111933000	14042
		HTTP/1.1 200 OK (text/html) 300		HTTP	192.168.8.117	34.223.124.45	45.112375500	14043
		Seq=514 Ack=1547 Win=65535 Len=0 [ACK] 80 → 57692 54		TCP	34.223.124.45	192.168.8.117	45.112430800	14044

0000	a0 29 42 b6 5f 0	42: Packet, 1354 bytes on wire (10832 bits), 1354 bytes captured (10832 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD005105F}, id 0 <
0010	05 3c c0 45 40 0	Ethernet II, Src: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74), Dst: Intel_b6:5f:03 (a0:29:42:b6:5f:03) <
0020	08 75 00 50 e1 5	Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.8.117 <
0030	68 b0 9d 45 00 0	Transmission Control Protocol, Src Port: 80, Dst Port: 57692, Seq: 1, Ack: 514, Len: 1300 >
0040	30 30 20 4f 4b 0	Source Port: 80
0050	2c 20 32 32 20 5	Destination Port: 57692
0060	3a 35 35 3a 31 3	[Stream index: 88]
0070	65 72 3a 20 41 7	[Stream Packet Number: 6]
0080	38 20 28 29 0d 0	[Conversation completeness: Complete, WITH_DATA (31)] <
0090	32 2c 68 32 63 0	[TCP Segment Len: 1300]
00a0	6e 3a 20 55 70 6	Sequence Number: 1 (relative sequence number)
00b0	2d 41 6c 69 76 6	Sequence Number (raw): 823472171
00c0	69 66 69 65 64 3	[Next Sequence Number: 1301 (relative sequence number)]
00d0	75 6e 20 32 30 3	Acknowledgment Number: 514 (relative ack number)
00e0	20 47 4d 54 0d 0	Acknowledgment number (raw): 3766064123
00f0	2d 35 65 32 38 6	Header Length: 20 bytes (5) = 0101
0100	7a 69 70 22 0d 0	Flags: 0x010 (ACK) <
0110	67 65 73 3a 20 6	Window: 26800
0120	3a 20 41 63 63 6	[Calculated window size: 26800]
0130	67 0d 0a 43 6f 6	
0140	69 6e 67 3a 20 6	
0150	6e 74 2d 4c 65 6	
0160	0a 4b 65 65 70 2	

Part 3: Capturing and Analyzing UDP Traffic

Task 2 – UDP Packet Details

	udp		Info	length	Protocol	Destination	Source	Time	No.
			Len=27 57790 → 443 69		UDP	192.168.8.117	216.239.38.120	3.579448900	8
			Len=29 443 → 57790 71		UDP	216.239.38.120	192.168.8.117	6.392151300	12
			Standard query 0x3bcf HTTP www.bing.com 92		DNS	_fe80::be3d:85ff:fed _fe80::3293:ded2:919	6.396032100	15	
			Standard query 0x3bcf A www.bing.com 92		DNS	_fe80::be3d:85ff:fed _fe80::3293:ded2:919	6.396320000	16	
			Standard query response 0x3bcf A www.bing.com CNAME www-wwww.bing.com.trafficmanager.net CNAME www 357		DNS	_fe80::3293:ded2:919 _fe80::be3d:85ff:fed	6.458003400	17	
			Standard query response 0x3bcf HTTPS www.bing.com CNAME www-wwww.bing.com.trafficmanager.net CNAME www 274		DNS	_fe80::3293:ded2:919 _fe80::be3d:85ff:fed	6.458471800	18	
			QUIC		QUIC	94.97.229.59	192.168.8.117	6.459872300	19
			QUIC		QUIC	94.97.229.59	192.168.8.117	6.459939200	20

0000

bc 3d 85 d4 78 7

19: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{817A0041-9545-4A82-9976-15ACD005105F}, id 0 <

0010

04 fe a5 68 40 0

Ethernet II, Src: Intel_b6:5f:03 (a0:29:42:b6:5f:03), Dst: HuaweiTechno_d4:78:74 (bc:3d:85:d4:78:74) <

0020

e5 3b f1 d7 01 b

Internet Protocol Version 4, Src: 192.168.8.117, Dst: 94.97.229.59 <

0030

19 7e 27 4f 7e c

User Datagram Protocol, Src Port: 61911, Dst Port: 443 >

0040

e5 68 07 00 6a b

Source Port: 61911

0050

da a5 aa 62 54 8

Destination Port: 443

0060

69 7b 1d 6e b6 7

Length: 1258

0070

d1 2c 5e d6 89 8

Checksum: 0x11b6 [unverified]

0080

df 7a e2 ad 92 e

[Checksum Status: Unverified]

0090

25 6d 35 cc 50 5

[Stream index: 3]

00a0

3b f8 4b 5b f3 8

[Stream Packet Number: 1]

00b0

8e fe 03 b6 fc 0

[Timestamps] <

00c0

f5 aa 00 19 f0 b

UDP payload (1258 bytes)

00d0

91 04 00 f7 dd 7

QUIC IETF <

00e0

dc 79 b0 cc 8a 3

00f0

95 fb 8c 47 7a a

0100

1d 3d fe 19 d8 d

0110

be 0d 95 6d 3c 2

0120

5a ha Ra h3 df 7

Part 4: Comparing TCP and UDP

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented: it establishes a session using the three-way handshake (SYN, SYN-ACK, ACK) before any data is sent, as observed in the Wireshark capture. It guarantees reliable delivery through acknowledgments (ACK numbers) and retransmits lost segments. UDP is connectionless — it sends datagrams directly with no handshake and no delivery guarantee, which is why the UDP capture showed no SYN/ACK exchange at all, only independent packets.
Data Integrity and Ordering	TCP	TCP uses Sequence and Acknowledgment numbers to keep segments in the correct order and detect missing or duplicated data, allowing correct reassembly of the byte stream (seen in the capture as "Seq=" / "Ack=" fields and "[TCP segment of a reassembled PDU]"). UDP has no sequence numbers or reassembly mechanism, so datagrams can arrive out of order, be duplicated, or be silently lost.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Applications that require guaranteed, ordered delivery, such as web browsing (HTTP/HTTPS), email, and file transfer. In the capture, the HTTP request to neverssl.com ran over TCP.	Applications that favor speed over reliability, such as live video/audio streaming, VoIP calls, online gaming, and DNS lookups. In the capture, DNS queries and YouTube's QUIC-based traffic ran over UDP.
Performance	Slower to start (handshake overhead) and has higher overhead per packet (larger header with sequence numbers, ACKs, flags, window size — 20+ bytes). Retransmissions and flow control add delay but ensure no data is lost.	Faster and lightweight: no handshake, minimal 8-byte header (source port, destination port, length, checksum), as seen in the capture. Lower latency, but no guarantee data arrives, arrives once, or arrives in order.